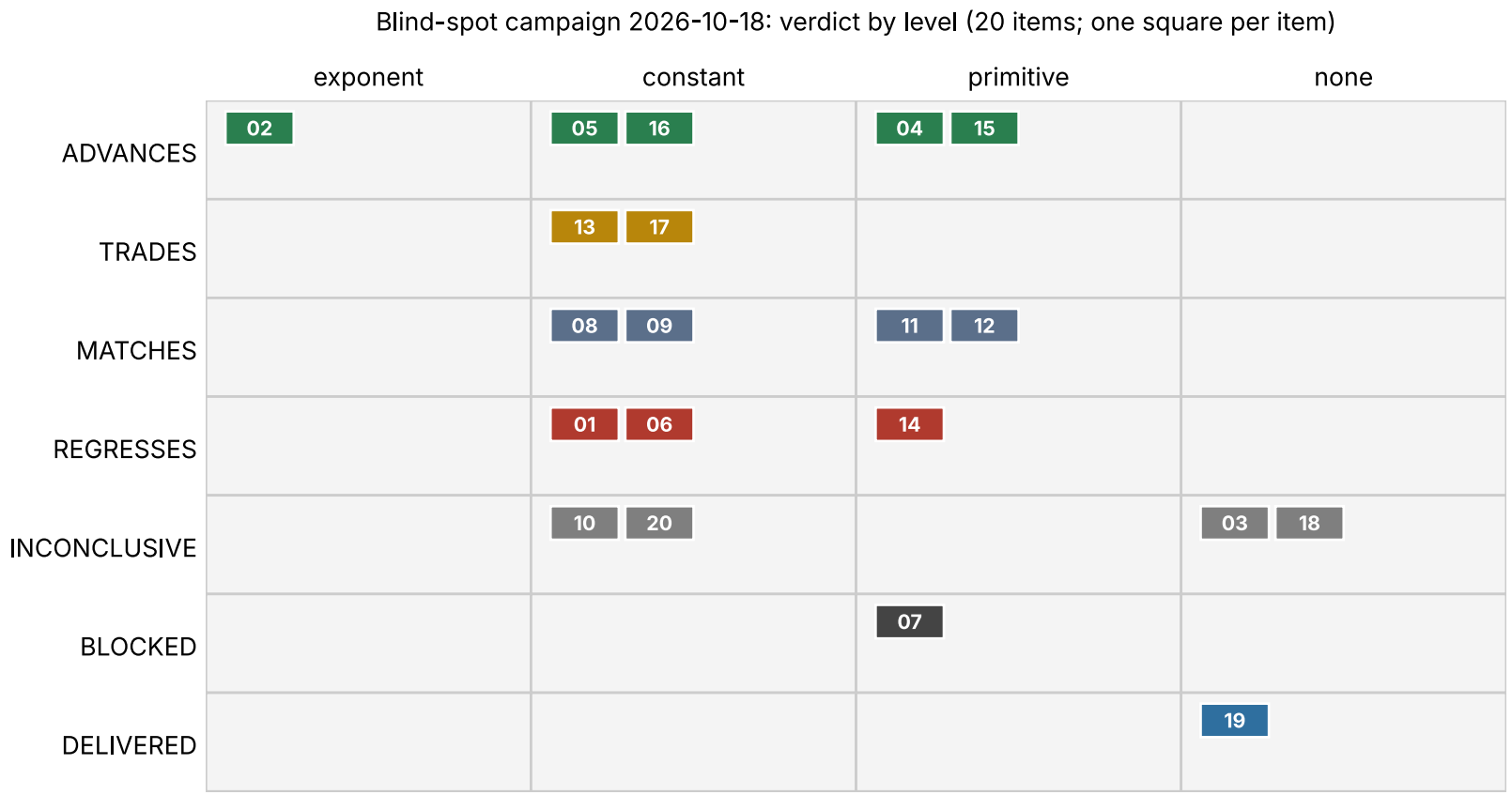
Blind-spot campaign, 2026-10-18: integration

Twenty experiments, one per direction the program was not pursuing (`EXPERIMENTS.md`). Each item ran under its own frozen `PROTOCOL.md` and decision rule; the verdict, level and numbers below are copied from the item README, which is the source of truth. Host for every item: shared 4-vCPU container, isolation L0, one thread per job. Every measured cell is toy scale.



Source: each item README verdict and level. Item 02 is a derived model; 03 and 18 state no level; all other levels are toy-scale stage diagnostics.

Graph source: `plot_campaign.py` (plots the verdict and level recorded in the item READMEs, no computation). PDF of this page: `REPORT.pdf`.

Table

item	title	verdict	level	headline	gap to rho	README
01	Gaudry k = 2 index calculus on E(F_{p^2})	REGRESSES	constant (exponent INCONCLUSIVE; plain-torsion ic-a REGRESSES at the exponent)	No arm beats matched rho at 24 to 40 bits or at any native-work price (0, 0.1, 1 GAE). Best arms ic-b and ic-c2 are 50x and 52x rho at w=0, 22x and 14x at w=1, at 40 bits.	9.9x at best (24 bits, w=1); 14x to 52x at 40 bits; extrapolated log2 gap at 246 bits about 5 to 17	01
02	Pricing binary curves at deployed sizes under the measured degree model	ADVANCES (derived, model level)	exponent (conditional on constant D)	Some m >= 4 design arms flip below matched rho at K=1 if the degree stays constant (D = 3, 4, 6, 7). The measured growth fit D = l+4 never flips. m=3	Koblitz m=4 w=1 K=1, D=6: +15.5 bits at n=131, −25.4 at n=571; D=l+4: +337 bits at n=571	02

				never flips; the m=2 control never crosses. Not a measured improvement.		
03	Polynomial-method solving on descended Semaev systems	INCONCLUSIVE	none (frozen sensitivity control failed: dinur moved 1.36% and 3.99% against a 1% bound)	Counted model at N=32: rho_dinur 0.0213 (46.8x worse than FES), rho_lptwy 0.0102 (97.7x worse). Fitted exponents FES 1.063, dinur 0.937, lptwy 0.967. Model drops below FES only at N=80 (needs 2^71 bits of memory) and N=128.	Exponent distance to matched rho 0.6946 (m=2), 0.6448 (m=3); no measured gap	03
04	Presentation search for the m-sum variety	ADVANCES (stage diagnostic, arm d)	primitive	Moebius-frame S_{m+1} with 2-torsion symmetrisation lowers closure degree D against the unchained incumbent on 5 of 5 cells (7 to 5, 7 to 5, 11 to 7, 10 to 7, >=13 to 10). It does not go below the chained S_3 route (D = 4); the symmetrisation itself lowers nothing (arm d0 gives the same D).	Not applicable (no cross-method row); unchanged	04
05	Monomial, variable and block orders for the descent	ADVANCES (arm wdeg only)	constant	wdeg cuts F4 matrix-entry touches against this engine's grevlex by 14.1x, 18.6x, 3.8x, 3.4x on four countable cells (N = 29 to 42), each above the cell's random-ranking maximum. D_mat is not lowered. At N=50 wdeg hit the cap while grevlex finished (reversal).	Not computed (stage diagnostic); unchanged	05
06	Signature-based Groebner engine with trace replay	REGRESSES	constant	sig is regress-certain in 3 of 4 cells (R = touches(f4)/cost(sig) at most 0.494, 0.497, 0.492); refuted 0 of 64 targets. Exact-key replay f4r has R 0.885, 0.630, 0.741, 0.587 and falls back on every target.	Unchanged	06
07	Out-of-core blocked Boolean F4 for degree cells	BLOCKED (time)	primitive	D = 4 on every completed draw up to (13,5,4); (15,5,4): one draw reads D > 4, two hit the time limit; (19,6,4) needs 22.97 GB. Cards 1 and 7 never completed.	Not computed	07
08	Learned branching for WDSat and XOR-CDCL	MATCHES	constant	No learned arm reaches 1.3x over the WDSat static order on 128 held-out targets (arm b 0.994, b2 1.001, c 1.000 at k=0). Noise-fitted scorers do as well.	Unchanged	08
09	Learned F4 selection and pivot policy	MATCHES (b50); d REGRESSES; c INCONCLUSIVE	constant	b50 touches ratio 1.99, 2.32, 2.54 against the incumbent but beats its random-half null on 1 cell of 3. Cell (13,3,4) trained, never evaluated.	Unchanged	09
10	Program search over the decomposition kernel	INCONCLUSIVE (cells: MATCHES n=31, ADVANCES n=37, ADVANCES n=41)	constant	Search finalist uses 11 to 12x fewer counted operations per relation in a replica, but about two-thirds of that gain comes from a one-gene lift patch (A1) and random search of equal budget finds much of the rest (Q(B)/ Q(C) 1.005, 0.628, 0.501).	Unchanged, not measured; a constant factor does not touch the exponent	10

11	Learned necessary-condition prefilter	MATCHES	primitive	No learned filter beats the trace filter by 5 points at zero false refutations; best credited 0.21 points against a 5-point threshold. Net work negative in all 16 cells.	Unchanged	11
12	Surrogate for the solving-degree surface	MATCHES	primitive	Boosted-tree surrogate beats the semi-regular reading (MAE 0.482 against 2.382) but is not separated from the engine-median reading (0.391); 90% intervals cover 73.5%.	Not applicable	12
13	Negation-map shortfall and ECC2K-130 plan	TRADES (part B DELIVERED)	constant	Of 44 collision-preserving settings: 0 ADVANCES, 30 TRADES, 2 MATCHES, 12 REGRESSES. Best fully charged gain 1.557 [1.325, 1.825] against sqrt2 = 1.414, with intervals including 1.35. Part B: 15.189 M iterations/s per core, 4,488 core-years CPU alone.	Gain over plain rho 1.0; best cells 1.49 to 1.56 with native work free; below 1.0 at 1 GAE	13
14	First prime-field arity m = 4 cost cells	REGRESSES	primitive	Arm (b) F4 cost grows as	F	^11.03 against the
15	Genus-3 linear-algebra exponent	ADVANCES (arm d only; exponent rule gives MATCHES)	primitive	Automatic SGE filter choice lowers LA cost 1.4% at b=20 (ratio 0.9862 [0.9836, 0.9876]), the size of the A/A spread (0.9875 to 1.011). No arm lowers the LA exponent; arms b and c REGRESS, e TRADES.	Whole IC cost 0.46 to 0.93 of one measured rho run per curve at b=20 (heavy-tailed rho, toy q about 2^20); exponent about 0.05 to 0.07 above 4/3	15
16	Size-gated equivariant oracle	ADVANCES (substitute barrel-oracle cells; production cells BLOCKED)	constant	Gated/incumbent CPU 2.4465 [2.3950, 2.5009] pooled; n=11 3.44x, n=13 10.40x. Most of the gain at n >= 11 is fall-row augmentation (F), not equivariance (adds about 1.13x and 1.34x). The gate is dead code in the production pipeline.	Not applicable (within-method)	16
17	Cyclotomic base cost arm at n = 83	TRADES	constant	Cyclotomic base is never cheaper than the compact-orbit-scan: REGRESSES at K=74 and 93, order flips with unpriced-work price at K=300; all arms within 0.11% under tidy accounting. Solving-degree leg BLOCKED.	232x to 1e9x above the rho theory floor (model rows, not run)	17
18	Execute the frozen ECDLP contracts	INCONCLUSIVE (no level; nothing compared)	none	3 of 1029 contracts runnable, 1026 BLOCKED; 4 replica trials run as frozen, all exit 0. Not a comparison, nothing improves. Lives in the sibling repository crypto-autoresearcher (commit eb4eedef8).	Not applicable	`crypto-autoresearcher/research/blindspots_20261018/18-frozen-contracts/README.md`
19	Negative-results paper and Lean statements	DELIVERED	none	Paper and evidence ledger delivered; S1 RECORDED, S2 PROVED (Lean unchecked), S3 PROVED and EXACT, S4 DERIVED UNREVIEWED, S5 EXACT, S6 NOT ESTABLISHED. A	Not applicable	19

				relabelling of existing negatives, not an improvement.		
20	Classical pre- and post-processing for quantum ECDLP	INCONCLUSIVE	constant	Negative control failed in 21 of 567 single-run cells, all at m=12 (post hoc: the chance formula is a union bound). Row-level outputs at n=256: E1-2^30 and E1-2^40 ADVANCE (0.927, 0.887; the latter sits at the 2^40 cap), lattice REGRESSES (1.715, 2.205). Logical-resource model, not a measured circuit.	Classical post-processing 70.0 to 120.9 bits remaining against rho at 127.83 bits (derived)	20

Counts: ADVANCES 5 (02, 04, 05, 15, 16), TRADES 2 (13, 17), MATCHES 4 (08, 09, 11, 12), REGRESSES 3 (01, 06, 14), INCONCLUSIVE 4 (03, 10, 18, 20), BLOCKED 1 (07), DELIVERED 1 (19).

What changed in the picture

Only what the evidence supports:

- **No whole-pipeline row moved and no gap to matched rho was reduced by a measurement.** Every cross-method measurement in the campaign is on the wrong side of rho: item 01 (14x to 52x at 40 bits), item 14 (6.5x to 30x, growing with p), item 17 (232x or more above the theory floor).
- **Item 02 is a model, not a measurement.** If the solving degree stays constant at l of 90 to 140, binary m >= 4 designs flip below matched rho for n above about 220 to 320 (K=1, w=1). The measured degree growth fit (D = l+4) never flips. The degree evidence is measured only to n <= 15, l <= 5, and item 07, which targets the deciding degree cells, is BLOCKED at (15,5,4). The question of whether D stays bounded is therefore still open.
- **Component gains exist, at toy scale, against the campaign's own baselines, and none is established as large.** Item 05 (wdeg touches 3.4x to 18.6x, reversed at N=50), item 04 (closure degree lowered against the unchained incumbent, not below the chained route), item 16 (2.4x pooled, mostly fall-row augmentation, production cells blocked), item 10 (about 11x in a replica, mostly a patch and random search) and item 15 (1.4%, within the A/A spread). None changes an exponent.
- **Learned components found nothing over their incumbents** (items 08, 09, 11, 12): the noise-fitted or random-subset controls match the learned arms.
- **Signature Groebner (06) and the cyclotomic base (17) do not help.** Negation-map rho (13) gains at most about 1.1 of sqrt2 with free native work and loses once native work is priced at 1 GAE.
- **Item 03:** the frozen sensitivity control failed, so no level is stated; the counted model puts the polynomial-method arms behind FES at N=32. Whether to relax the 1% bound is an owner decision.
- Items 18 and 19 and the INCONCLUSIVE item 20 are bookkeeping or model results and claim no improvement.

Under the success criterion (every verified, measured improvement over the index-calculus baseline counts), the candidates for that label are items 05, 16, 04 and 10, each as a toy-scale, within-method stage measurement with the limits above. This page claims none as an improvement of the baseline.

Blocked and unfinished items, and what unblocks each

item	blocked	what unblocks it
03	Frozen sensitivity control failed (1.36% and 3.99% against 1%; the code accepts under 5%); arm (e) msolve not installed	Owner decision on the 1% bound; install msolve for the comparison arm; descriptive branch with the control waived is TRADES at exponent over constant
04	Cell 5 half measured: arm e BLOCKED(memory) (basis 2.91 GB against the 3000 MiB cap), arm b blocked by projection	A memory cap above 3 GB for arm e, or an out-of-core engine (item 07)
06	Arm (e) msolve not installed; cell c15-3-4 unsettled (F4 time limit on target 1); f4rs covers 1 to 4 targets per cell	msolve; a longer time limit for c15-3-4 target 1
07	BLOCKED(time) at (15,5,4); draw g0 missing; rank controls for (19,7,3) incomplete; (19,6,4) needs 22.97 GB, 16.05 GB beyond the cap	More wall time per job and a larger memory cap, or a disk-backed matrix; re-run cards 1 and 7
08	CryptoMiniSat arm descriptive only (6 targets); scale l=5	Larger l and more CMS targets
09	Cell (13,3,4) trained, never evaluated; random-subset clause undecided (about 1,570 s of budget left)	Freeze and run the fit and evaluation cards for that cell
14	Arm (d) BLOCKED by reach (delta_max 0.0081 against 0.25 needed); arm (b) at 20 to 28 bits stops at its count cap	A reachable base (two abscissae) design cell; a higher count cap for F4
16	Production cells n in {41,47,53,61} BLOCKED: 64-bit monomials against 170 to 250 unknowns, 192-bit masks against 211 to 311 symmetric unknowns, dense matrix 37 to 377 GB against the 4 GB cap, no Frobenius-symmetric system in the production pipeline, crypto_lib does not compile at the branch base	Wider monomial and mask types, a sparse or out-of-core matrix, a Frobenius-symmetric production system, a compiling library
17	Solving-degree leg BLOCKED: the m=2 system needs 166 or more variables against the 64-variable limit	A solver with more than 64 variables
18	1026 of 1029 contracts BLOCKED (first gates: not_frozen:status 353, no_launcher 285, has_runs 186, withheld 147, not_frozen:no_specification 29, handoff_archive_coverage 9, needs_dispatch_queue 6)	Freeze the specifications, write launchers, record approvals, per contract; the supervisor path was not taken (replica records only)

19	Lean not kernel-checked (no toolchain, about 1.5 GB disk); S4 has no independent review; S6 open; authors and email owed by a human	The CI proof gate (`lean-proof-gate.yml`); an independent proof review
10	Card 22-r2 ran without a card or receipt	A frozen card for the summarize step

Leaderboard

No item qualifies as a large measured gain on an index-calculus pipeline, phase or component, so `docs/ic-leaderboard.html`, `docs/ic/LEADERBOARD.md`, `docs/ic/leaderboard.json` and the lab browser were not regenerated. Reasons: item 05 is a toy-scale touch count against the engine's own grevlex that reverses at N=50 and does not lower D; item 16's production cells are blocked and its gain is mostly fall-row augmentation; item 04 lowers a closure degree, not a cost; item 10 is a replica whose gain is mostly a patch and random search; item 15's 1.4% is within the A/A spread; item 02 is derived. Items 05 and 16 are the closest; an owner who counts them as component speedups would add them as their own section outside tables A to C (AGENTS.md section 7b).

Benchmarks

Twelve items copied files into `benchmarks/20261018-blindspots-*/` (01, 02, 03, 05, 06, 07, 10, 13, 14, 15, 16, 17), and each has its row in `benchmarks/README.md`. Items 04, 08, 09, 11, 12, 18, 19 and 20 are not speed results and have no copy.

Process caveats shared by the items

- Many relay checks report `relay_verbatim` false, or `CARD FAIL`, because frozen cards name outputs that exist only after the run; items record this rather than repairing frozen cards. Receipts and on-disk outputs are the evidence.
- Several items were restarted or repaired once; each README keeps its first run under "First run (superseded)".
- Conductor is not installed in the container, so no `conductor check` was possible.